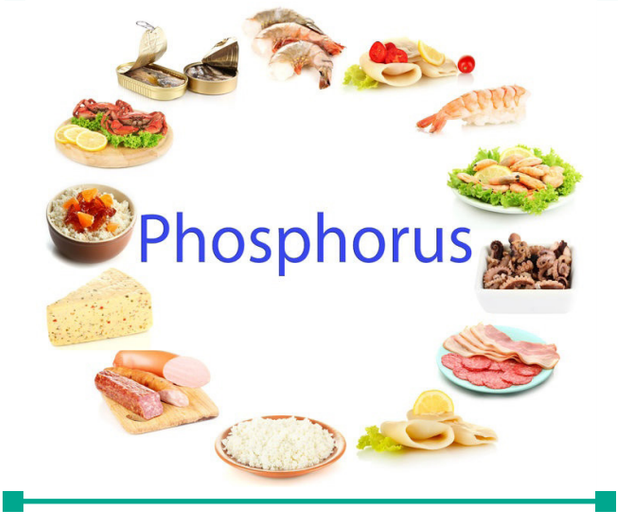
The difference between osteopenia and osteoporosis is a matter of severity of the loss of bone density.

- **Tetany:** A decrease in serum calcium levels gives rise to a condition called tetany. The symptoms of tetany are severe intermittent spasms of the muscles of hands and feet accompanied by muscular pain. Twitching of facial muscles occurs.

(ii) Phosphorus

- **Distribution:** It comprises 1 percent of total body weight along with calcium. An adult human body contains about 400-700 g of phosphorus as phosphates. Bones and teeth contain 85 percent of phosphorus and soft tissues contain15 percent of phosphorus.
- **Functions:**
  - Formation of bone and teeth along with calcium and magnesium.
  - Formation of phospholipids which are integral parts of cell structure.
  - Constituent of co enzymes like coenzyme I and co – carboxylase.
  - Integral constituent of DNA and RNA (nucleic acids) and nucleoproteins.
  - Buffering of acid or alkali excesses to maintain normal pH.
  - Temporary storage and transfer of the energy derived from metabolic fuels.
  - As part of enzymes needed for the metabolism of carbohydrates, protein and fats.
- **Food Sources:** Phosphorus is widely distributed in foods. Milk and meat are rich in phosphorus. Whole grain cereals, legumes, nuts, carrots and fish are also rich sources of phosphorus.



- **Calcium Phosphorus ratio:** Nutritionists recommend that a Ca:P ratio **between 1:1 to 2:1** should be provided by the total diet.
- **Deficiency:** Phosphorus is so ubiquitous in various foods that near total starvation is required to produce dietary phosphorus deficiency. Inadequate phosphorus intake is expressed as hypophosphatemia which manifests in the form of anemia, anorexia, muscle weakness, bone pain, rickets, osteomalacia, general weakness and increased susceptibility to infection.

(iii) Iron

- **Distribution:** Iron content of normal adult man is estimated to be about 4 grams. Iron is distributed as 60% in the circulating **hemoglobin**,

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